SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science & Engineering

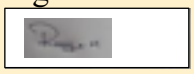
ASSESSMENT TOOL 1 – CONCEPT MAPPING

Course Code	**CSA0315**	**Course Title**	**Data Structures**
CO Assessed	**CO1**	**Assessment**	**Assessment Tool 1: Concept Mapping**
Weightage	**35% of CIA**	**Total Marks**	**100**
CO4 Statement	***Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity.* (BL2, BL3)**		
Student Name	**RAMYA M**	**Reg. No.**	**1921260148**
Section/Batch	**Slot D/2026**	**Date**	**05-10-2026**

Code of Conduct:

Iam, Ramya M certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts	
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established	
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure	
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence	
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand	

SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

SIMATS ENGINEERING
Engineer to Excel

CONCEPT MAP

BIG DATA SYSTEM

(Millions Of Records)

01 PERFORMANCE

Overall speed and responsiveness of the big data system.

KEY FACTOR:

- Response time
- latency

02 TIME COMPLEXITY

Measures how the processing time increases with data size.

EXAMPLE:

- O(n)
- O(log n)

03 SPACE COMPLEXITY

Measures the memory/storage required for processing

EXAMPLE:

- O(n)
- O(log n)

04 THROUGHPUT

Number of records processed in a given time

MEASURED IN:

- second
- MB/s
- GB/s

05 EFFICIENCY

Achieves more output using fewer time and memory resources.

FACTORS:

- Cost
- Scalability

06 OPTIMIZATION

Improves algorithms, memory usage, parallel processing and data handling.

STRATEGIES:

Efficient algorithms

IMPACT ON SYSTEM DESIGN

- Guides choice of algorithms and data structures
- Helps set resource allocation (CPU, memory, storage)
- Influences parallel processing and distributed architecture
- Ensures scalability and handles large data volumes effectively

REG NO:1921260148